Botswana International University of Science & Technology

Faculty of Engineering and Technology

Department of Electronics, Computer and Telecommunication Engineering

End of Semester One Examinations 2022/2023

Course Code: CPEN 522

Course or Module Title: Robotics Modelling and Simulations

Date & time of Examination: 0800hrs – 1200hrs, 24th November 2022

Time allowed: 2 hours

Total Marks: 100

The following items are provided:

One Examination Paper (3 printed pages in total inclusive of cover page)

Candidates are permitted to bring into the examination room:

- Pen, pencil and eraser

The paper consists of 4 QUESTIONS, Answer ALL FOUR QUESTIONS. Each question carries a total mark indicated.

- All answers must be written in the answer book(s) provided.
- All rough work must be written in the answer book(s) provided.
- A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be graded.
- Black or Blue pen only must be used for written answers and pencil for all drawings and sketches.
- Ensure you have the correct number of pages on your examination question paper.

Question 1 [20 marks]

a) Define the below:

(i) Sensors
(ii) End-Effector
(iii) Degree of freedom
(iv) Kinematics
(v) Kinetics **[10 marks]**

b) List and explain any 3 parts of a robot. **[6 marks]**

c] List 2 advantages and 2 disadvantages of a robot. **[4 marks]**

Question 2 [20 marks]

a) The polar coordinates of a point are (-5, 0.23). Determine the Cartesian Coordinates for the point. **[5 marks]**

b) The Cartesian coordinated of a point are (-8, 1). Determine the polar coordinates of this point. **[5 marks]**

c) Write a program for the system that regulates the speed of a blinking LED with a potentiometer. **[10 marks]**

Question 3 [25 marks]

a. Outline the workspace properties. **[4 marks]**

b. Using appropriate functions and symbols define POSE. **[4 marks]**

c. Based on the source of input power actuators are classified in to three groups, outline these groups, providing each group's advantages and limitations. **[6 marks]**

d. For a rotary actuator, determine
   i. Angular velocity, ω,
   ii. and Torque, T.

   [6 marks]

e. Draw the following circuits and provide a brief description on how they work.
   i. Temperature Sensor
   ii. Light Sensor

   [5 marks]

Question 4 [35 marks]

a. In terms of robotic trajectory planning, define in detail the following terms
   i. Path
   ii. Trajectory
   iii. Joint space description
   iv. Cartesian space description

[8 marks]

b. Consider a simple 2 degree of freedom robot. We desire to move the robot from Point A to Point B, see Figure 1, below. Let's assume that both joints of the robot can move at the maximum rate of 10 degree/sec. Initial angles $\alpha = 20^{o}$ and $\beta = 30^{o}$

[10 marks]

1. Describe the robot arm movement in terms of links:
   i. Angular velocities
   ii. Timing e.g., after 2 seconds and after 3 seconds
   iii. Path and distances travelled
   iv. Angles α and β at each point

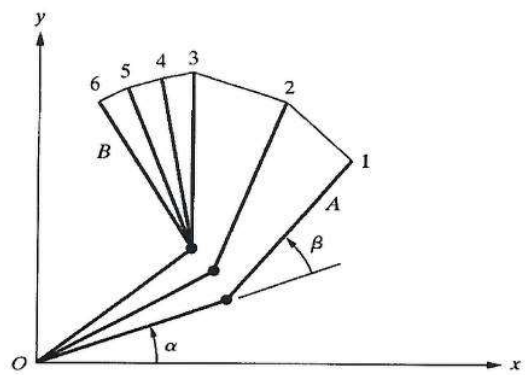
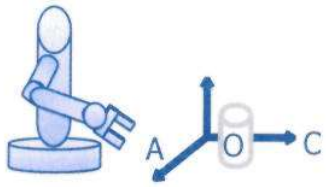


Figure 1: Joint-space nonnormalized movements of a robot with two degrees of freedom

2. Assume that the robot's hand follows a known path between points A and B in a straight line.
   i. Outline the method that could be used for trajectory planning
   ii. Draw the cartesian-space movements of the two-degree-of-freedom robot
   iii. Provide angles α and β at each point
   iv. Describe the steps for trajectory planning with an acceleration-deceleration

[12 marks]

c. Plan a Cartesian trajectory from A to C (rest-to-rest) that avoids the obstacle O, with $a \leq a_{\max}$ *and* $v \leq v_{\max}$



[5 marks]